Institute of Technology, Nirma University

Computer Networks

24BTM020

Practical 1

Practical 1 Tasks

1. PING Command

Objective

To test connectivity between two hosts and measure network response time.

Exercise Questions

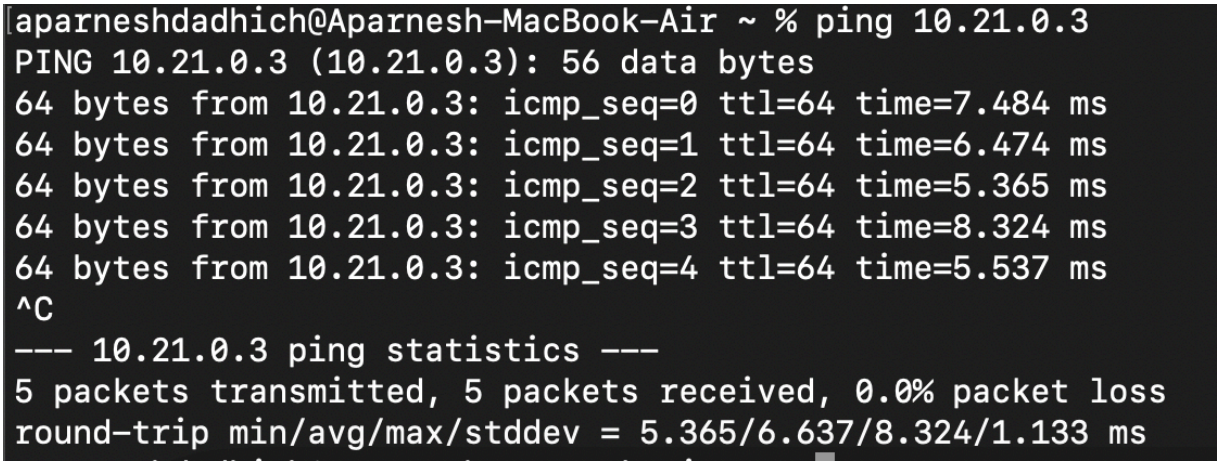
1. Write the syntax of the ping command in Windows and Linux.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % ping
usage: ping [-AaDdfnoQqRrv] [-c count] [-G sweepmaxsize]
            [-g sweepminsize] [-h sweepincrsize] [-i wait]
            [-l preload] [-M mask | time] [-m ttl] [-p pattern]
            [-S src_addr] [-s packetsize] [-t timeout][-W waittime]
            [-z tos] host
       ping [-AaDdfLnoQqRrv] [-c count] [-I iface] [-i wait]
            [-l preload] [-M mask | time] [-m ttl] [-p pattern] [-S src_addr]
            [-s packetsize] [-T ttl] [-t timeout] [-W waittime]
            [-z tos] mcast-group
```

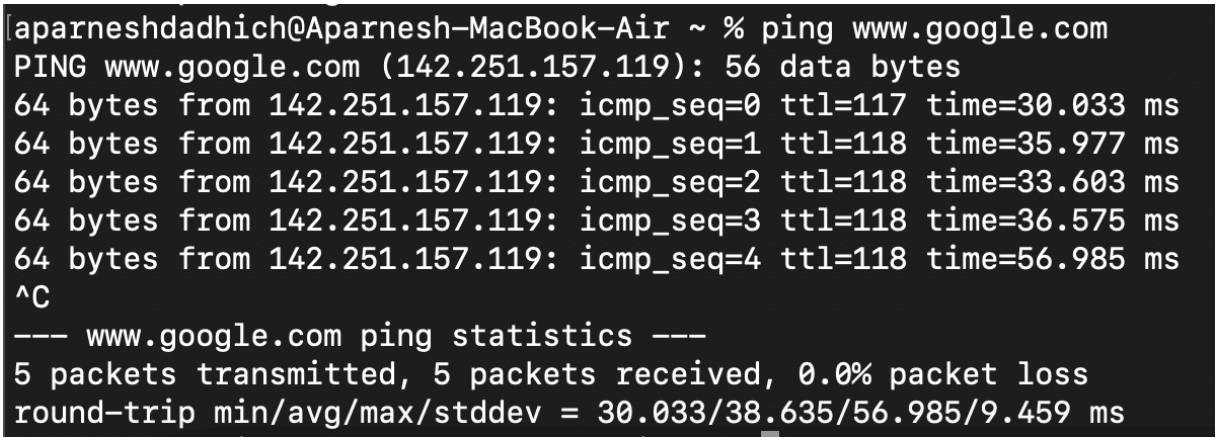
2. Ping the localhost and record the observations.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % ping localhost
PING localhost (127.0.0.1): 56 data bytes
64 bytes from 127.0.0.1: icmp_seq=0 ttl=64 time=0.086 ms
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.170 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.202 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.152 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.148 ms
^C
--- localhost ping statistics ---
5 packets transmitted, 5 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.086/0.152/0.202/0.038 ms
```

3. Ping your default gateway and note the packet loss and RTT.



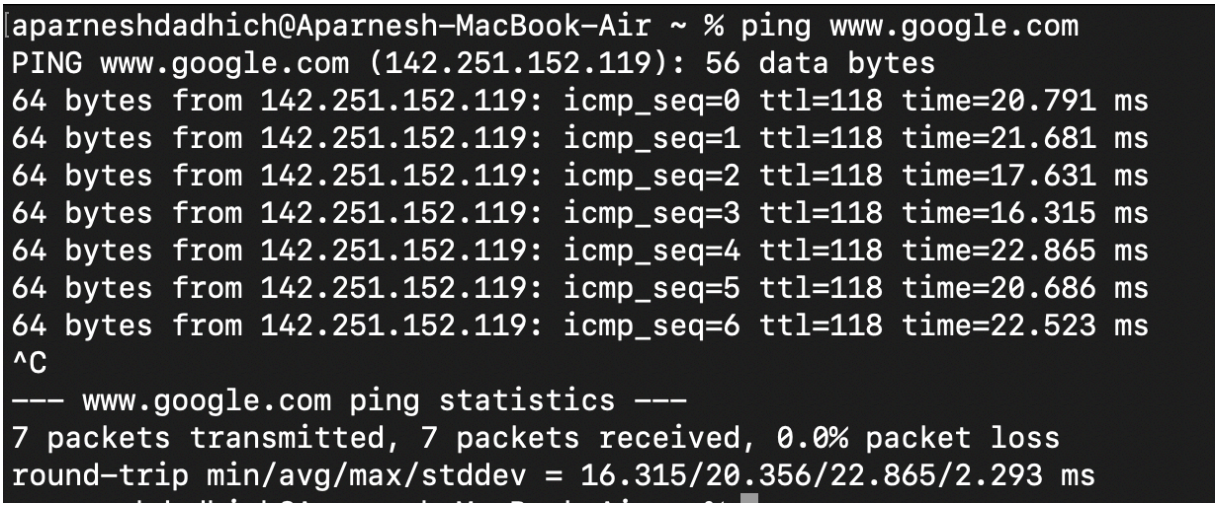
4. Ping the domain google.com. What does the command verify?



5. Explain the significance of TTL in ping results.

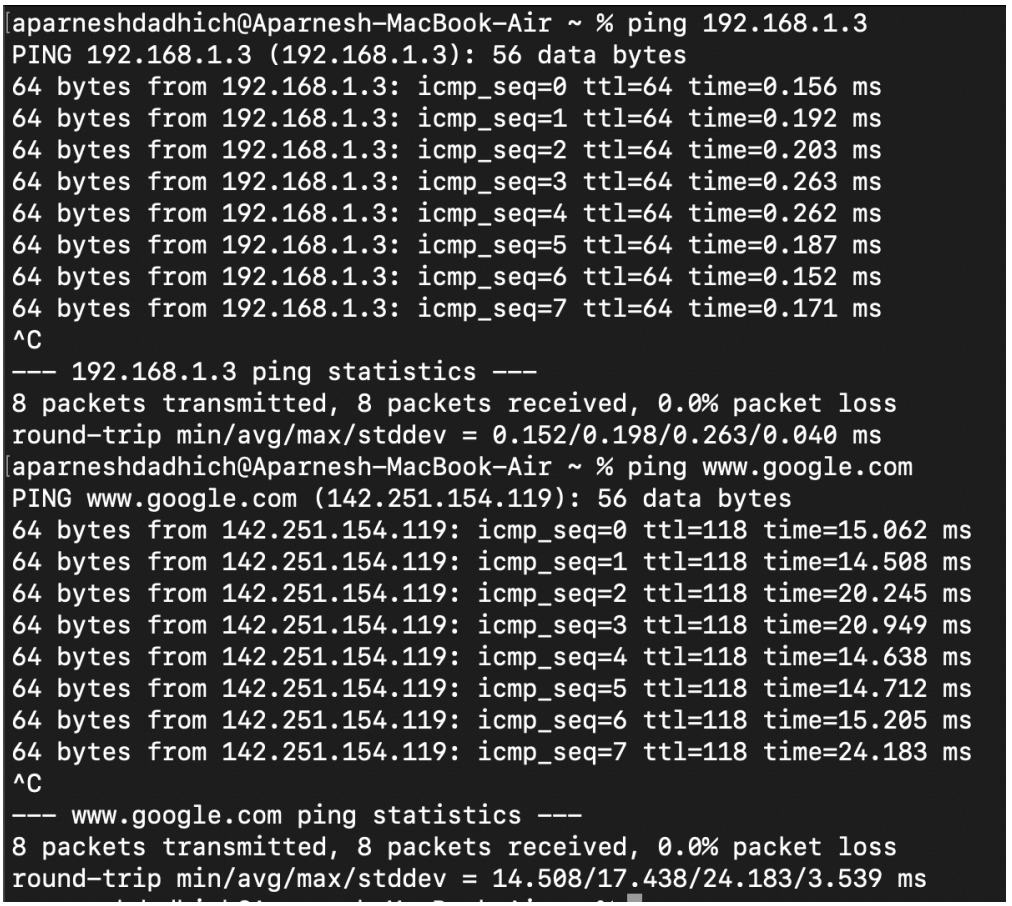
TTL (Time To Live) is an IP packet field that limits the number of routers a packet can pass through. Each router decrements the TTL by 1, and if it reaches 0, the packet is discarded. This prevents packets from looping indefinitely in the network. In ping results, the TTL value indicates that the destination host responded successfully and provides an estimate of the number of network hops between the source and destination.

6. Execute continuous ping and stop it manually. Write the command used.



Ctrl + C is used to stop the ping command manually.

7. Compare ping results for a local IP address and a remote website.



Local IP addresses have very low response times because the communication stays within the local network. Remote websites have higher response times due to Internet routing and longer

transmission distances.

8. What happens when you ping an unreachable host?

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % ping 192.168.1.250
PING 192.168.1.250 (192.168.1.250): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3
ping: sendto: No route to host
Request timeout for icmp_seq 4
ping: sendto: Host is down
Request timeout for icmp_seq 5
ping: sendto: Host is down
Request timeout for icmp_seq 6
ping: sendto: Host is down
Request timeout for icmp_seq 7
ping: sendto: Host is down
Request timeout for icmp_seq 8
ping: sendto: Host is down
Request timeout for icmp_seq 9
```

9. Use ping with packet size modification and record the output.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % ping -s 100 google.com
PING google.com (142.251.222.142): 100 data bytes
108 bytes from 142.251.222.142: icmp_seq=0 ttl=118 time=42.809 ms
108 bytes from 142.251.222.142: icmp_seq=1 ttl=118 time=42.079 ms
108 bytes from 142.251.222.142: icmp_seq=2 ttl=118 time=41.715 ms
108 bytes from 142.251.222.142: icmp_seq=3 ttl=118 time=42.519 ms
108 bytes from 142.251.222.142: icmp_seq=4 ttl=118 time=42.450 ms
108 bytes from 142.251.222.142: icmp_seq=5 ttl=118 time=41.700 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 41.700/42.212/42.809/0.415 ms
```

10. Differentiate between ICMP Echo Request and Echo Reply.

An ICMP Echo Request is sent by the source device to check if a host is reachable, while an ICMP Echo Reply is the response sent by the destination to confirm it is reachable.

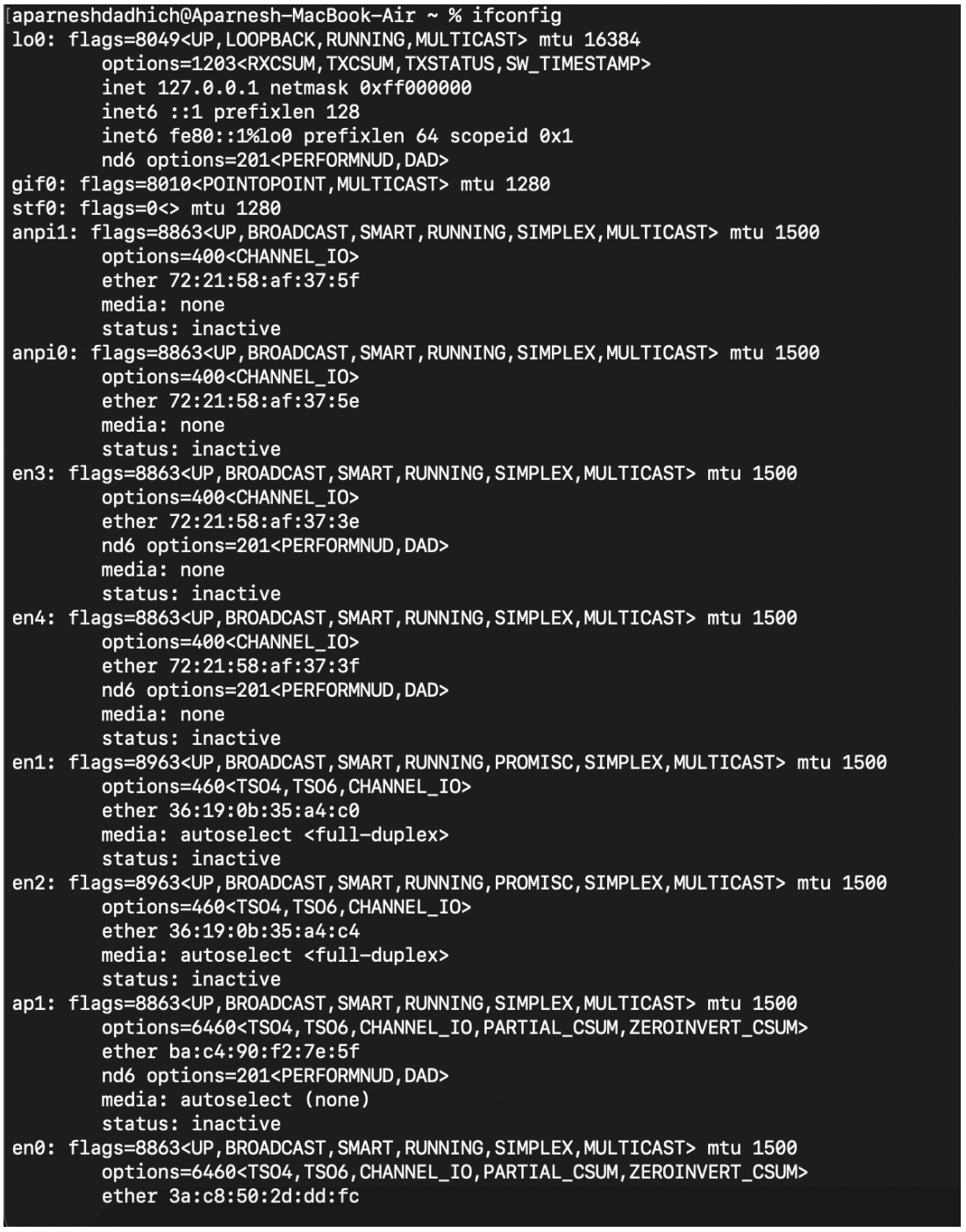
2. IPCONFIG / IFCONFIG Command

Objective

To view and analyze network interface configuration.

Exercise Questions

1. Display the IP configuration of your system using ipconfig or ifconfig.



```
aparneshdadhich@Aparnesh-MacBook-Air ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
        options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
        inet 127.0.0.1 netmask 0xff000000
        inet6 ::1 prefixlen 128
        inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
        nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 72:21:58:af:37:5f
        media: none
        status: inactive
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 72:21:58:af:37:5e
        media: none
        status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 72:21:58:af:37:3e
        nd6 options=201<PERFORMNUD,DAD>
        media: none
        status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 72:21:58:af:37:3f
        nd6 options=201<PERFORMNUD,DAD>
        media: none
        status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:19:0b:35:a4:c0
        media: autoselect <full-duplex>
        status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:19:0b:35:a4:c4
        media: autoselect <full-duplex>
        status: inactive
ap1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether ba:c4:90:f2:7e:5f
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect (none)
        status: inactive
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 3a:c8:50:2d:dd:fc
```

```
        ether 3a:c8:50:2d:dd:fc
        inet6 fe80::1c0f:d43b:f80c:5555%en0 prefixlen 64 secured scopeid 0xb
        inet6 2401:4900:1f3f:dcbe:10d6:607c:3cec:58d9 prefixlen 64 autoconf secured
        inet6 2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9 prefixlen 64 autoconf temporary
        inet 192.168.1.3 netmask 0xffffff00 broadcast 192.168.1.255
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=63<RXCSUM,TXCSUM,TSO4,TSO6>
        ether 36:19:0b:35:a4:c0
        Configuration:
                id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
                maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
                root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
                ipfilter disabled flags 0x0
        member: en1 flags=3<LEARNING,DISCOVER>
                ifmaxaddr 0 port 8 priority 0 path cost 0
        member: en2 flags=3<LEARNING,DISCOVER>
                ifmaxaddr 0 port 9 priority 0 path cost 0
        nd6 options=201<PERFORMNUD,DAD>
        media: <unknown type>
        status: inactive
awdl0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 5a:57:71:31:7d:54
        inet6 fe80::5857:71ff:fe31:7d54%awdl0 prefixlen 64 scopeid 0xd
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
llw0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 5a:57:71:31:7d:54
        inet6 fe80::5857:71ff:fe31:7d54%llw0 prefixlen 64 scopeid 0xe
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect (none)
utun0: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
        inet6 fe80::c90c:c8f2:f34b:c288%utun0 prefixlen 64 scopeid 0xf
        nd6 options=201<PERFORMNUD,DAD>
utun1: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1500
        inet6 fe80::5173:ae5e:fdd3:1666%utun1 prefixlen 64 scopeid 0x10
        nd6 options=201<PERFORMNUD,DAD>
utun2: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000
        inet6 fe80::9172:7d1d:b0ef:d1d6%utun2 prefixlen 64 scopeid 0x11
        nd6 options=201<PERFORMNUD,DAD>
utun3: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000
        inet6 fe80::ce81:b1c:bd2c:69e%utun3 prefixlen 64 scopeid 0x12
        nd6 options=201<PERFORMNUD,DAD>
```

2. Identify:
    - **IPv4 Address:** A unique 32-bit logical address assigned to a device for communication on an IP network. It identifies the device and is used to send and receive data (e.g., 192.168.1.10).
    - **Subnet Mask:** A 32-bit value that separates the network portion and host portion of an IPv4 address. It helps determine whether a destination device is on the same network or a different one.
    - **Default Gateway:** The IP address of the router that forwards data from the local

network to other networks, such as the Internet. It is used when the destination is outside the local subnet.

- **MAC Address:** A unique 48-bit physical address assigned to a network interface by the manufacturer. It is used for communication between devices within the same local network.

3. Explain the purpose of DHCP.

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and other network settings, such as the subnet mask, default gateway, and DNS server, to devices on a network. It eliminates manual IP configuration and helps prevent IP address conflicts.

4. Use ipconfig /all and list additional information obtained.

Additional information: ipconfig /all displays detailed network information such as the host name, MAC address, DHCP status, DNS servers, DHCP server, lease details, and IPv6 configuration.

5. Release and renew the IP address using appropriate commands.

```
[aparneshdadhich@Aparnesh-MacBook-Air ~ % sudo ipconfig set en0 NONE
[aparneshdadhich@Aparnesh-MacBook-Air ~ % sudo ipconfig set en0 DHCP
```

These commands release the current network configuration and request a new IP address from the DHCP server.

6. Differentiate between static and dynamic IP addressing.

Static IP Address: A static IP is manually assigned to a device and remains the same unless changed manually.

Dynamic IP Address: A dynamic IP is automatically assigned by a DHCP server and may change each time the device reconnects to the network.

7. What is the purpose of loopback address?

The loopback address (127.0.0.1) is used to test the local network stack and allows a device to communicate with itself without using a physical network.

8. Compare outputs of ifconfig and ip addr.

ifconfig displays basic network interface information such as IP and MAC addresses, while ip addr provides more detailed and modern network configuration information.

9. Identify active and inactive interfaces.

- **Active interface:** An interface that is **UP/RUNNING** and has a valid IP address

assigned (e.g., en0, wlan0).

Inactive interface: An interface that is **DOWN** or not connected, and may not have an IP address assigned.

10. Explain the significance of subnet mask in communication.

The subnet mask identifies the network and host portions of an IP address, helping devices determine whether the destination is on the same network or should be reached through a router.

3. IWCONFIG Command

Objective
To examine wireless network interface parameters.

Exercise Questions

1. Display wireless interface details using iwconfig.

iwconfig <interface>

2. Identify the ESSID and operating frequency

The ESSID is the name of the connected wireless network (Wi-Fi name), and the operating frequency is the radio frequency used by the network, typically 2.4 GHz or 5 GHz.

3. What is the significance of signal level and link quality?

- Signal level indicates the strength of the received Wi-Fi signal. It is measured in dBm.
- Link quality is the overall quality of the connection, not just the signal strength. It also reflects factors like interference, noise, and packet loss. You can have a strong signal but poor link quality if there's a lot of interference from other Wi-Fi networks or devices.

4. Explain Managed and Monitor modes.

Managed mode is the normal mode where a device connects to a Wi-Fi network. The device only processes data meant for your connection.

Monitor mode allows the wireless adapter to capture all nearby Wi-Fi packets without connecting to any network. It's used for network analysis, troubleshooting, and

security testing.

5. Change the wireless mode using iwconfig.

To change the wireless mode on a Linux system, the network interface must first be turned down, switched to the desired mode, and then brought back up. The core structural command used for the mode alteration itself is iwconfig [interface_name] mode [managed/monitor].

6. Observe bitrate and transmission power values.

- Bitrate: The maximum speed at which data is currently being transmitted over the wireless connection, measured in Megabits per second (Mbps). It drops automatically as you move further away from the router.
- Transmission Power (Tx-Power): The amount of electrical power the wireless card uses to broadcast its radio signal, measured in dBm or milliwatts. Higher power increases range but consumes more battery.

7. Compare wired and wireless interface configurations.

A **wired interface** (Ethernet) connects using an Ethernet cable and is configured with settings like IP address, subnet mask, gateway, MAC address, and DNS. It provide more stable, faster connections.

A **wireless interface** (Wi-Fi) uses the same network settings but also includes Wi-Fi-specific parameters such as SSID, operating frequency, signal strength, link quality, and security type (WPA2/WPA3).

8. Explain the purpose of Access Point MAC address.

The **Access Point MAC address** is the unique hardware address of the wireless router or access point, used to identify it and enable devices to connect to the correct network.

9. What happens when ESSID is off/any?

When the **ESSID** is off/any, the access point does not broadcast its network name, making the Wi-Fi network hidden. Users must manually enter the ESSID to connect.

10. List limitations of wireless communication observed from the output.

Wireless communication has limitations such as limited range, signal interference from other devices or Wi-Fi networks, and obstacles like walls or furniture that weaken the signal. As a result, signal strength and link quality may fluctuate, leading to reduced speed, packet loss, or unstable connections.

4. NETSTAT Command

Objective

To inspect active network connections and ports.

Exercise Questions

1. Display all active connections using netstat.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % netstat -an
Active Internet connections (including servers)
Proto Recv-Q Send-Q  Local Address          Foreign Address        (state)
tcp4       0      0  192.168.1.3.62068      104.20.16.140.443      ESTABLISHED
tcp4       0      0  192.168.1.3.62065      3.94.72.89.443         ESTABLISHED
tcp4       0      0  192.168.1.3.62064      3.94.72.89.443         ESTABLISHED
tcp4       0      0  192.168.1.3.62062      20.42.65.94.443        ESTABLISHED
tcp4       0      0  192.168.1.3.62061      20.42.65.94.443        ESTABLISHED
tcp4      63      0  192.168.1.3.62045      54.205.56.37.443       CLOSE_WAIT
tcp4       0      0  192.168.1.3.62026      172.66.43.16.443       ESTABLISHED
tcp4       0      0  192.168.1.3.62021      108.158.46.111.443     ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.62000 2600:1901:1:388:.443   ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.61969 2404:6800:4000:1.443   ESTABLISHED
tcp4       0      0  192.168.1.3.61955      98.90.71.17.443        ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.61872 2600:1901:1:388:.443   ESTABLISHED
tcp4       0      0  192.168.1.3.61802      52.23.108.170.443      ESTABLISHED
tcp4       0      0  192.168.1.3.61683      172.188.155.25.443     ESTABLISHED
tcp4       0      0  192.168.1.3.61614      100.59.129.106.443     ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.61605 2404:6800:4007:8.443   ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.61478 2404:6800:4007:8.443   ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.61356 2600:1901:1:7c5:.443   ESTABLISHED
tcp4       0      0  192.168.1.3.61343      52.23.78.155.443       ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.61342 2603:1040:a06:6:.443   ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.61340 2404:6800:4000:1.443   ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.61338 2404:6800:4000:1.443   ESTABLISHED
tcp4       0      0  *.61335                *.*                    LISTEN
tcp6       0      0  2401:4900:1f3f:d.58781 2600:1901:1:d18:.443   ESTABLISHED
tcp4       0      0  192.168.1.3.58761      104.199.241.202.4070   ESTABLISHED
tcp6       0      0  *.58728                *.*                    LISTEN
tcp4       0      0  *.58728                *.*                    LISTEN
tcp4       0      0  *.57621                *.*                    LISTEN
tcp4       0      0  127.0.0.1.7768         *.*                    LISTEN
tcp46      0      0  *.51861                *.*                    LISTEN
tcp6       0      0  *.5000                 *.*                    LISTEN
tcp4       0      0  *.5000                 *.*                    LISTEN
tcp6       0      0  *.7000                 *.*                    LISTEN
tcp4       0      0  *.7000                 *.*                    LISTEN
tcp4       0      0  127.0.0.1.58273        *.*                    LISTEN
tcp4       0      0  *.88                   *.*                    LISTEN
tcp6       0      0  *.88                   *.*                    LISTEN
tcp4       0      0  *.5900                 *.*                    LISTEN
tcp6       0      0  *.5900                 *.*                    LISTEN
tcp4       0      0  192.168.1.3.62060      192.178.211.84.443     TIME_WAIT
tcp4       0      0  192.168.1.3.51014      104.26.12.49.443       ESTABLISHED
tcp4       0      0  192.168.1.3.51013      104.26.12.49.443       ESTABLISHED
tcp4       0      0  192.168.1.3.51012      104.26.12.49.443       ESTABLISHED
tcp4       0      0  192.168.1.3.51011      104.26.12.49.443       ESTABLISHED
tcp4       0      0  192.168.1.3.51010      104.26.12.49.443       TIME_WAIT
tcp6       0      0  2401:4900:1f3f:d.63267 2606:4700:90d2:7.443   ESTABLISHED
tcp6       0      0  2401:4900:1f3f:d.63205 2606:4700:90d2:7.443   ESTABLISHED
```

2. Identify listening ports on the system.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % netstat -l
Active Internet connections
Proto Recv-Q Send-Q  Local Address                                 Foreign Address                                 (state)
tcp4       0      0  192.168.1.3.62084                            ec2-34-199-66-18.https                          ESTABLISHED
tcp4       0      0  192.168.1.3.62083                            ec2-34-199-66-18.https                          ESTABLISHED
tcp4       0      0  192.168.1.3.62068                            104.20.16.140.https                             ESTABLISHED
tcp4       0      0  192.168.1.3.62065                            ec2-3-94-72-89.c.https                          ESTABLISHED
tcp4       0      0  192.168.1.3.62064                            ec2-3-94-72-89.c.https                          ESTABLISHED
tcp4       0      0  192.168.1.3.62062                            20.42.65.94.https                               ESTABLISHED
tcp4       0      0  192.168.1.3.62061                            20.42.65.94.https                               ESTABLISHED
tcp4       0      0  192.168.1.3.62026                            172.66.43.16.https                              ESTABLISHED
tcp4       0      0  192.168.1.3.62021                            server-108-158-4.https                          ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.62000 2600:1901:1:388::.https                         ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.61872 2600:1901:1:388::.https                         ESTABLISHED
tcp4       0      0  192.168.1.3.61683                            172.188.155.25.https                            ESTABLISHED
tcp4       0      0  192.168.1.3.61614                            ec2-100-59-129-1.https                          ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.61605 maa05s20-in-x0a.1e100.net.https                 ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.61478 pnmaaa-au-in-x05.1e100.net.https                ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.61356 2600:1901:1:7c5::.https                         ESTABLISHED
tcp4       0     35  192.168.1.3.61343                            ec2-52-23-78-155.https                          ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.61342 2603:1040:a06:6::2.https                        ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.61340 pt-in-f95.1e100.net.https                       ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.61338 pt-in-f95.1e100.net.https                       ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.58781 2600:1901:1:d18::.https                         ESTABLISHED
tcp4       0      0  192.168.1.3.58761                            202.241.199.104..4070                           ESTABLISHED
tcp4       0      0  192.168.1.3.51015                            104.26.12.49.443                                ESTABLISHED
tcp4       0      0  192.168.1.3.51014                            104.26.12.49.443                                ESTABLISHED
tcp4       0      0  192.168.1.3.51013                            104.26.12.49.443                                ESTABLISHED
tcp4       0      0  192.168.1.3.51012                            104.26.12.49.443                                ESTABLISHED
tcp4       0      0  192.168.1.3.51011                            104.26.12.49.443                                TIME_WAIT
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.63267 2606:4700:90d2:7cbc:c2ca:4d5:5ff2:75c4.443      ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.63205 2606:4700:90d2:7cbc:c2a7:4d5:5ff2:75c4.443      ESTABLISHED
tcp4       0      0  192.168.1.3.50913                            ec2-63-176-195-2.443                            ESTABLISHED
tcp4       0      0  192.168.1.3.50900                            104.26.12.49.443                                ESTABLISHED
tcp4       0      0  192.168.1.3.50896                            17.242.13.5.5223                                ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.63130 whatsapp-chatd-edge6-shv-01-bom5.facebo.443     ESTABLISHED
tcp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.63064 2600:1901:1:d18::.443                           ESTABLISHED
udp46      0      0  *.57211                                      *.*
udp46      0      0  *.62071                                      *.*
udp46      0      0  *.57650                                      *.*
udp46      0      0  *.49642                                      *.*
udp6       0      0  2401:4900:1f3f:dcbe:5d00:5a3f:1da:ebc9.56806 pnmaaa-an-in-x0e.1e100.net.https
udp4       0      0  *.57621                                      *.*
udp4       0      0  *.61532                                      *.*
udp4       0      0  192.168.1.3.63370                            unn-152-233-9-12.commplex-link
udp4       0      0  192.168.1.3.61897                            unn-149-88-31-12.commplex-link
udp6       0      0  *.mdns                                       *.*
udp6       0      0  *.mdns                                       *.*
udp6       0      0  *.mdns                                       *.*
udp4       0      0  *.mdns                                       *.*
```

3. Explain the meaning of TCP states:
    - **LISTENING:** The port is open and waiting for incoming connection requests from remote devices. The server is ready to accept new connections.
    - **ESTABLISHED:** A TCP connection has been successfully created between two devices, and data can be sent and received.
    - **TIME_WAIT:** The connection has been closed, but the system waits for a short time to ensure delayed packets are received and to prevent connection conflicts before reusing the port.

4. Find the process ID associated with a port.

```
TCP    10.237.102.149:49215   104.154.127.247:4070   ESTABLISHED     19604
```

5. Display routing table using netstat options.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % netstat -r
Routing tables

Internet:
Destination        Gateway            Flags               Netif Expire
default            192.168.1.1        UGScg                 en0
127                localhost          UCS                   lo0
localhost          localhost          UH                    lo0
169.254            link#11            UCS                   en0      !
192.168.1          link#11            UCS                   en0      !
192.168.1.1/32     link#11            UCS                   en0      !
192.168.1.1        6c:4f:89:ac:d7:28  UHLWIir               en0   1174
192.168.1.3/32     link#11            UCS                   en0      !
192.168.1.5        32:b:75:90:bb:50   UHLWIi                en0    929
192.168.1.6        b2:e3:5:cc:71:ec   UHLWI                 en0    558
192.168.1.255      ff:ff:ff:ff:ff:ff  UHLWbI                en0      !
224.0.0/4          link#11            UmCS                  en0      !
mdns.mcast.net     1:0:5e:0:0:fb      UHmLWI                en0
239.255.255.250    1:0:5e:7f:ff:fa    UHmLWI                en0
255.255.255.255/32 link#11            UCS                   en0      !

Internet6:
Destination        Gateway            Flags               Netif Expire
default            fe80::6e4f:89ff:fe UGcg                  en0
default            fe80::%utun0       UGcIg               utun0
default            fe80::%utun1       UGcIg               utun1
default            fe80::%utun2       UGcIg               utun2
default            fe80::%utun3       UGcIg               utun3
localhost          localhost          UHL                   lo0
2401:4900:1f3f:dcb link#11            UC                    en0
2401:4900:1f3f:dcb 3a:c8:50:2d:dd:fc  UHL                   lo0
2401:4900:1f3f:dcb 3a:c8:50:2d:dd:fc  UHL                   lo0
fe80::%lo0         aparnesh-macbook-a UcI                   lo0
aparnesh-macbook-a link#1             UHLI                  lo0
fe80::%en0         link#11            UCI                   en0
aparnesh-macbook-a 3a:c8:50:2d:dd:fc  UHLI                  lo0
fe80::1cef:9eb1:d7 32:b:75:90:bb:50   UHLWI                 en0
fe80::6e4f:89ff:fe 6c:4f:89:ac:d7:28  UHLWIir               en0
fe80::5857:71ff:fe 5a:57:71:31:7d:54  UHLI                  lo0
fe80::5857:71ff:fe 5a:57:71:31:7d:54  UHLI                  lo0
fe80::%utun0       aparnesh-macbook-a UcI               utun0
aparnesh-macbook-a link#15            UHLI                  lo0
fe80::%utun1       aparnesh-macbook-a UcI               utun1
aparnesh-macbook-a link#16            UHLI                  lo0
fe80::%utun2       aparnesh-macbook-a UcI               utun2
aparnesh-macbook-a link#17            UHLI                  lo0
fe80::%utun3       aparnesh-macbook-a UcI               utun3
aparnesh-macbook-a link#18            UHLI                  lo0
ff00::             localhost          UmCI                  lo0
ff00::             link#11            UmCI                  en0
```

6. Compare TCP and UDP entries.

- TCP entries show connection-oriented communication and include connection states such as LISTEN, ESTABLISHED, and TIME_WAIT, since TCP establishes a reliable connection before transmitting data. It is used for reliable connection. It is used for faster communication where reliability is not required.

- UDP entries are connectionless and do not have connection states. They simply show the local address and port where the application is sending or receiving datagrams.

7. Identify foreign and local addresses.

   Local Address: Represents the IP address and port number of your own computer involved in the network connection. It identifies the local endpoint of the communication.

   Foreign Address: Represents the IP address and port number of the remote device or server connected to your system. It identifies the remote endpoint.

8. Explain why some ports remain in TIME_WAIT state.

   Ports remain in the TIME_WAIT state for a short period after a TCP connection is closed to ensure all delayed or duplicate packets are discarded. This also allows the final acknowledgment (ACK) to be retransmitted if needed, preventing conflicts with future connections using the same port.

9. Display statistics of network protocols.

   netstat -s displays statistics for network protocols such as TCP, UDP, IP, and ICMP. It shows the number of packets sent and received, errors, retransmissions, and other protocol-specific statistics.

10. What security risks can be identified using netstat?

netstat can help identify suspicious or unauthorized network connections, unknown listening ports, and unexpected remote connections that may indicate malware or unauthorized access. It can also reveal unnecessary open ports and services that could be exploited by attackers.

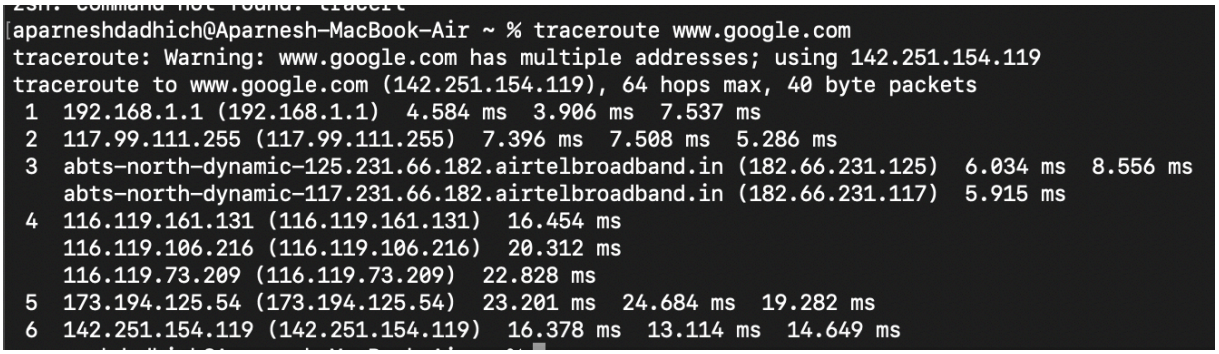
5. TRACEROUTE / TRACERT Command

Objective

To determine the path taken by packets to a destination.

Exercise Questions

1. Trace the route to google.com.



2. Record the number of hops observed.

Number of hops observed = 6

3. Explain the significance of each hop.

- Hop 1 (192.168.1.1): Your local router (default gateway) that forwards packets from your device to the ISP.
- Hop 2 (117.99.111.255): The first router in your Internet Service Provider's (ISP) network.
- Hop 3 (Airtel Broadband routers): Intermediate ISP routers that route traffic within the Airtel network.
- Hop 4 (116.119.x.x): Backbone/transit routers that forward packets toward Google's network.
- Hop 5 (173.194.125.54): A router within or very close to Google's network.
- Hop 6 (142.251.154.119): The destination server (www.google.com) successfully reached.

4. Why do some hops display * * *?

Some hops display * * * because the router does not respond to traceroute requests or blocks ICMP packets due to firewall or security settings.

5. Compare ping and traceroute commands.

Ping is used to check whether a destination host is reachable and measures the round-trip time (RTT) of packets.

Traceroute identifies the path taken by packets by displaying each intermediate router (hop) between the source and destination along with the delay at each hop. In short, ping verifies connectivity, while traceroute helps diagnose where network delays or failures occur.

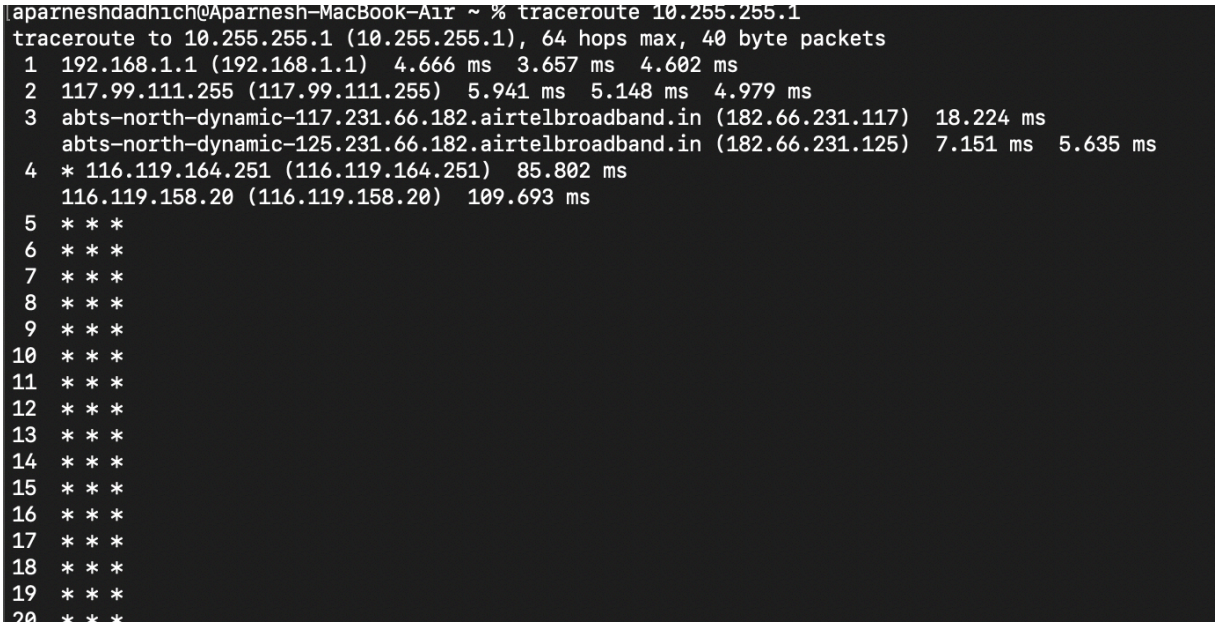
6. Explain the role of TTL in traceroute.

TTL (Time to Live) is a field in the IP packet that limits how many routers (hops) the packet can pass through. Traceroute sends packets with increasing TTL values (1, 2, 3, …). When the TTL reaches zero, the router discards the packet and sends back an ICMP “Time Exceeded” message, allowing traceroute to identify each hop along the path.

7. Identify the ISP router from the output.

The ISP router is Hop 2 (117.99.111.255), which is the first router belonging to your Internet Service Provider after your local router. The following hops (3 and 4), identified as Airtel Broadband routers, are also part of the ISP’s network.

8. Trace a route to an unreachable destination.



9. Measure delay variation between hops.

Delay variation between hops is observed by comparing the three response times shown for each hop in the traceroute output. Small variations are normal due to changing network traffic, while large variations may indicate congestion or network delays.

10. Explain practical applications of traceroute.

Traceroute is used to identify the path packets take to reach a destination and locate where network problems occur. It helps diagnose high latency, packet loss, and routing issues between networks. Network administrators also use it to troubleshoot connectivity problems and verify network routes.

6. ROUTE Command

Objective

To view and manipulate routing tables.

Exercise Questions

1. Display the routing table using route print or route.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % route get default
   route to: default
destination: default
       mask: default
    gateway: 10.237.102.233
  interface: en0
      flags: <UP,GATEWAY,DONE,STATIC,PRCLONING,GLOBAL>
 recvpipe  sendpipe  ssthresh  rtt,msec    rttvar  hopcount      mtu     expire
       0         0         0         0         0         0      1500         0
```

2. Identify the default route in the table.

3. Explain:
   - **Destination:** The destination network or host that the route applies to. When a packet is sent, the system checks this field to determine if the route matches the packet's destination IP address.
   - **Gateway:** The IP address of the next-hop router that forwards packets toward the destination network. If the destination is on the local network, this field may indicate that no gateway is required.
   - **Genmask:** Defines the size of the destination network by specifying which bits of the IP address represent the network portion. It helps the system decide whether an IP address belongs to a particular route.
   - **Metric:** A value that represents the cost or priority of a route. When multiple routes to the same destination exist, the route with the **lowest metric** is usually chosen because it is considered the most efficient.
4. Add a temporary route using the route command.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % sudo route -n add -net 192.168.2.0/24 192.168.1.1
Password:
add net 192.168.2.0: gateway 192.168.1.1
```

5. Delete a route from the routing table

.

```
[aparneshdadhich@Aparnesh-MacBook-Air ~ % sudo route -n delete -net 192.168.2.0/24 192.168.1.1
delete net 192.168.2.0: gateway 192.168.1.1
```

6. Differentiate between static and dynamic routing.

Static routing requires routes to be manually configured and updated by the network administrator, making it suitable for small and stable networks.

Dynamic routing uses routing protocols such as RIP, OSPF, or EIGRP to automatically learn and update routes when the network changes. Static routing is simpler and more secure but less flexible, while dynamic routing is more scalable and adapts automatically to network failures or topology changes.

7. Explain the significance of routing metrics.

Routing metrics are values used by routing protocols to determine the best path for forwarding packets. They may consider factors such as hop count, bandwidth, delay, reliability, or cost. When multiple routes are available, the route with the lowest metric is usually selected as the most efficient path.

8. Observe changes after adding a route.

After adding a route, a new entry appears in the routing table with the specified destination, gateway, and subnet mask. Traffic destined for that network is then forwarded through the newly added route until it is deleted or the system is restarted.

9. What is the role of gateway in routing?

The **gateway** acts as the next-hop router that forwards packets from the local network toward their destination. It enables devices to communicate with networks outside their own subnet, including the internet.

10. Explain how routing tables help packet forwarding.

A routing table stores information about available network paths, including destinations, gateways, and metrics. When a packet is sent, the system checks the routing table to choose the best route and forwards the packet through the appropriate interface or gateway.

7. NSLOOKUP Command

Objective

To perform DNS queries and domain name resolution.

Exercise Questions

1. Find the IP address of google.com.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % nslookup google.com
Server:         2401:4900:ae3b:44c3::57
Address:        2401:4900:ae3b:44c3::57#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.174.139
Name:   google.com
Address: 192.178.174.100
Name:   google.com
Address: 192.178.174.138
Name:   google.com
Address: 192.178.174.101
Name:   google.com
Address: 192.178.174.113
Name:   google.com
Address: 192.178.174.102
```

2. Query the mail server records of a domain.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % nslookup -type=MX google.com
Server:         2401:4900:ae3b:44c3::57
Address:        2401:4900:ae3b:44c3::57#53

Non-authoritative answer:
google.com      mail exchanger = 10 smtp.google.com.

Authoritative answers can be found from:
smtp.google.com internet address = 192.178.211.26
smtp.google.com internet address = 192.178.211.27
smtp.google.com has AAAA address 2404:6800:4000:1025::1b
smtp.google.com has AAAA address 2404:6800:4000:1025::1a
```

3. Explain forward and reverse DNS lookup.

Forward DNS Lookup: Converts a domain name to an IP address (e.g., google.com to 142.250.x.x).

Reverse DNS Lookup: Converts an IP address to its domain name (e.g., 142.250.x.x to google.com).

4. Find the authoritative DNS server of a domain.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % nslookup -type=NS google.com
Server:         2401:4900:ae3b:44c3::57
Address:        2401:4900:ae3b:44c3::57#53

Non-authoritative answer:
google.com      nameserver = ns3.google.com.
google.com      nameserver = ns2.google.com.
google.com      nameserver = ns1.google.com.
google.com      nameserver = ns4.google.com.

Authoritative answers can be found from:
ns3.google.com  internet address = 216.239.36.10
ns3.google.com  has AAAA address 2001:4860:4802:36::a
ns2.google.com  internet address = 216.239.34.10
ns2.google.com  has AAAA address 2001:4860:4802:34::a
ns1.google.com  internet address = 216.239.32.10
ns1.google.com  has AAAA address 2001:4860:4802:32::a
ns4.google.com  internet address = 216.239.38.10
ns4.google.com  has AAAA address 2001:4860:4802:38::a
```

5. Compare nslookup and ping outputs.

nslookup resolves a domain name to its IP address and displays DNS information.

ping checks connectivity to the destination and measures response time using ICMP Echo Requests.

6. Query using a specific DNS server.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % nslookup google.com 8.8.8.8
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.126.102
Name:   google.com
Address: 142.251.126.138
Name:   google.com
Address: 142.251.126.101
Name:   google.com
Address: 142.251.126.100
Name:   google.com
Address: 142.251.126.139
Name:   google.com
Address: 142.251.126.113
```

7. What happens when DNS resolution fails?

   If DNS resolution fails, the domain name cannot be converted into an IP address, so the website or network service cannot be reached using its hostname.

8. Explain the purpose of DNS caching.

   DNS caching stores previously resolved domain names and their IP addresses, reducing lookup time and network traffic, which speeds up access to websites.

9. Perform reverse lookup for a given IP address.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % nslookup 8.8.8.8
Server:         2401:4900:ae3b:44c3::57
Address:        2401:4900:ae3b:44c3::57#53

Non-authoritative answer:
8.8.8.8.in-addr.arpa    name = dns.google.

Authoritative answers can be found from:
```

10. Identify different DNS record types.

- **A** - Maps a domain name to an IPv4 address.
- **AAAA** - Maps a domain name to an IPv6 address.
- **MX** - Specifies the mail server for a domain.
- **NS** - Identifies the authoritative name servers.
- **CNAME** - Creates an alias for another domain name.
- **PTR** - Used for reverse DNS lookups (IP to domain).

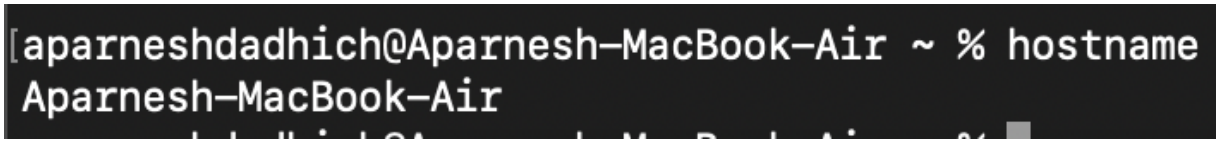
8. HOSTNAME Command

Objective

To identify and manage system hostnames.

Exercise Questions

1. Display the hostname of your machine.



2. Explain the role of hostname in networking.

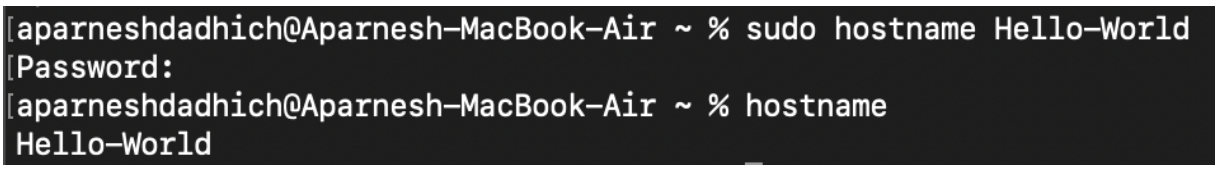
   A hostname provides a human-readable name for a device on a network, making it easier to identify and communicate with instead of using IP addresses.

3. Differentiate hostname and domain name.

   A **hostname** is the name assigned to a specific device on a network. A hostname usually represents an individual machine (e.g., server1).

   Domain name identifies a website or organization on the Internet. It is a registered Internet address (e.g., google.com).

4. Change the hostname temporarily.



5. Verify hostname resolution using ping.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % ping Aparnesh-MacBook-Air.local
PING aparnesh-macbook-air.local (127.0.0.1): 56 data bytes
64 bytes from 127.0.0.1: icmp_seq=0 ttl=64 time=0.075 ms
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.182 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.163 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.178 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.168 ms
^C
--- aparnesh-macbook-air.local ping statistics ---
5 packets transmitted, 5 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.075/0.153/0.182/0.040 ms
```

6. Explain FQDN (Fully Qualified Domain Name).

A **Fully Qualified Domain Name (FQDN)** is the complete and unique domain name of a host on the Internet, specifying its hostname and domain hierarchy. For example, www.google.com is an FQDN, where www is the hostname and google.com is the domain name.

7. How does hostname assist in network administration?

Hostnames make it easier for network administrators to identify, manage, and access devices without remembering IP addresses. They also simplify troubleshooting, monitoring, and remote administration of networked systems.

8. Observe hostname entry in configuration files.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % scutil --get HostName
Aparnesh-MacBook-Air
aparneshdadhich@Aparnesh-MacBook-Air ~ % scutil --get LocalHostName
Aparnesh-MacBook-Air
aparneshdadhich@Aparnesh-MacBook-Air ~ % scutil --get ComputerName
Aparnesh's MacBook Air
```

9. What issues arise from duplicate hostnames?

Duplicate hostnames can cause device identification conflicts, making it difficult to distinguish systems on a network. They may also lead to DNS resolution issues, connection errors, and problems with remote access and network management.

10. Compare hostname usage in Windows and Linux.

Windows and Linux both use hostnames to identify devices on a network. In Windows, the hostname is managed through System Settings or the hostname

command, while in Linux it is managed using commands like hostname or hostnamectl and configuration files such as /etc/hostname.

9. DIG Command

Objective

To perform detailed DNS queries.

Exercise Questions

1. Query A records of a domain using dig.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % dig google.com A

; <<>> DiG 9.10.6 <<>> google.com A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 64071
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             241     IN      A       192.178.211.100
google.com.             241     IN      A       192.178.211.138
google.com.             241     IN      A       192.178.211.102
google.com.             241     IN      A       192.178.211.113
google.com.             241     IN      A       192.178.211.101
google.com.             241     IN      A       192.178.211.139

;; Query time: 69 msec
;; SERVER: 2401:4900:ae3b:44c3::57#53(2401:4900:ae3b:44c3::57)
;; WHEN: Tue Jul 21 09:15:15 IST 2026
;; MSG SIZE  rcvd: 135
```

2. Query MX records and explain their purpose.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % dig google.com MX

; <<>> DiG 9.10.6 <<>> google.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 3885
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 5

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;google.com.                    IN      MX

;; ANSWER SECTION:
google.com.             300     IN      MX      10 smtp.google.com.

;; ADDITIONAL SECTION:
smtp.google.com.        300     IN      A       192.178.211.26
smtp.google.com.        300     IN      A       192.178.211.27
smtp.google.com.        300     IN      AAAA    2404:6800:4000:1025::1b
smtp.google.com.        300     IN      AAAA    2404:6800:4000:1025::1a

;; Query time: 125 msec
;; SERVER: 2401:4900:ae3b:44c3::57#53(2401:4900:ae3b:44c3::57)
;; WHEN: Tue Jul 21 09:15:57 IST 2026
;; MSG SIZE  rcvd: 148
```

3. Compare outputs of dig and nslookup.

dig provides detailed DNS query information, including headers, TTL, and response sections, making it useful for troubleshooting and analysis.

nslookup gives a simpler, easier-to-read output that is mainly used for basic DNS lookups and verification.

4. Perform reverse DNS lookup using dig.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % dig -x 8.8.8.8

; <<>> DiG 9.10.6 <<>> -x 8.8.8.8
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25833
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;8.8.8.8.in-addr.arpa.          IN      PTR

;; ANSWER SECTION:
8.8.8.8.in-addr.arpa.   19157   IN      PTR     dns.google.

;; Query time: 37 msec
;; SERVER: 2401:4900:ae3b:44c3::57#53(2401:4900:ae3b:44c3::57)
;; WHEN: Tue Jul 21 09:17:59 IST 2026
;; MSG SIZE  rcvd: 73
```

5. Identify TTL values in DNS responses.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % dig google.com

; <<>> DiG 9.10.6 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 59018
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             15      IN      A       192.178.211.100
google.com.             15      IN      A       192.178.211.138
google.com.             15      IN      A       192.178.211.102
google.com.             15      IN      A       192.178.211.113
google.com.             15      IN      A       192.178.211.101
google.com.             15      IN      A       192.178.211.139

;; Query time: 38 msec
;; SERVER: 2401:4900:ae3b:44c3::57#53(2401:4900:ae3b:44c3::57)
;; WHEN: Tue Jul 21 09:19:01 IST 2026
;; MSG SIZE  rcvd: 135
```

Here the TTL is 15.

6. Explain DNS header sections in dig output.

The DNS header in dig output contains information such as the query status (e.g., NOERROR), operation code (opcode), flags, and a unique ID. It also shows the number of entries in the QUERY, ANSWER, AUTHORITY, and ADDITIONAL sections of the DNS response.

7. Query NS records of a domain.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % dig google.com NS

; <<>> DiG 9.10.6 <<>> google.com NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 5846
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 9

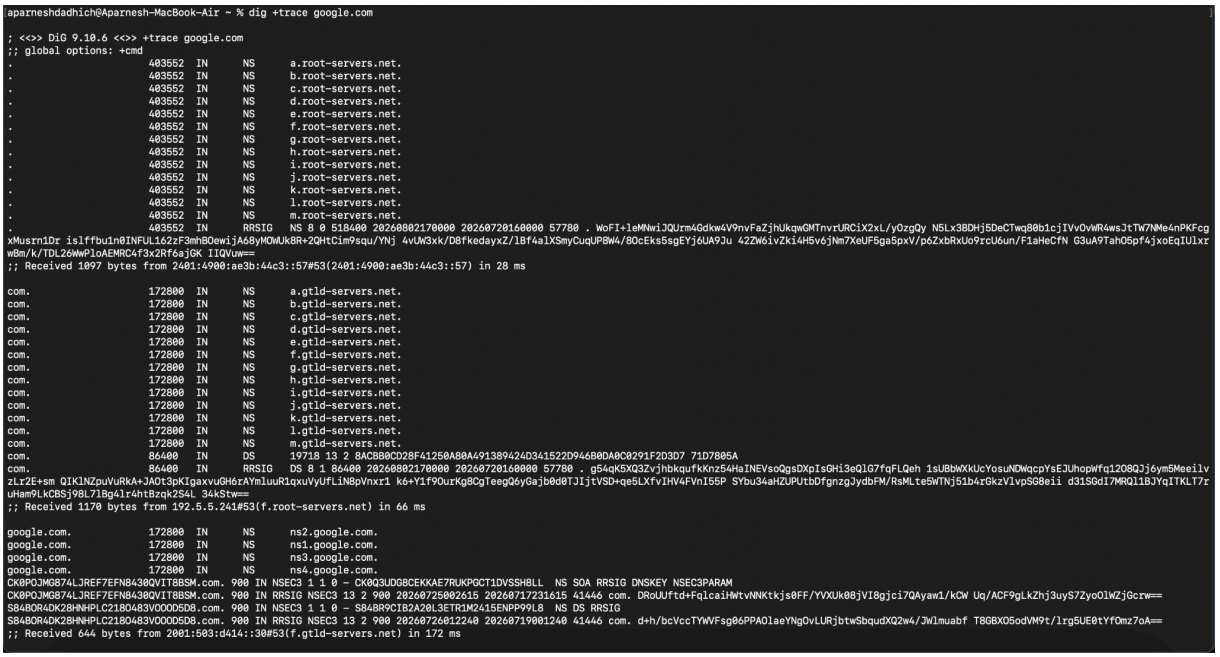
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;google.com.                    IN      NS

;; ANSWER SECTION:
google.com.             269140  IN      NS      ns2.google.com.
google.com.             269140  IN      NS      ns1.google.com.
google.com.             269140  IN      NS      ns3.google.com.
google.com.             269140  IN      NS      ns4.google.com.

;; ADDITIONAL SECTION:
ns2.google.com.         256067  IN      A       216.239.34.10
ns2.google.com.         256067  IN      AAAA    2001:4860:4802:34::a
ns1.google.com.         83087   IN      A       216.239.32.10
ns1.google.com.         83087   IN      AAAA    2001:4860:4802:32::a
ns3.google.com.         295809  IN      A       216.239.36.10
ns3.google.com.         295809  IN      AAAA    2001:4860:4802:36::a
ns4.google.com.         130451  IN      A       216.239.38.10
ns4.google.com.         130451  IN      AAAA    2001:4860:4802:38::a

;; Query time: 64 msec
;; SERVER: 2401:4900:ae3b:44c3::57#53(2401:4900:ae3b:44c3::57)
;; WHEN: Tue Jul 21 09:21:54 IST 2026
;; MSG SIZE  rcvd: 287
```

8. Trace DNS resolution using dig options.



9. What is recursion in DNS queries?

   Recursion in DNS means a DNS resolver performs all necessary queries to other DNS servers on behalf of the client until it finds the requested IP address. The client receives the final answer without contacting multiple DNS servers itself.

10. Explain practical applications of dig.

    dig is commonly used to troubleshoot DNS issues, verify DNS records, and test DNS server responses. It is also used by network administrators to diagnose domain resolution problems and validate DNS configurations.

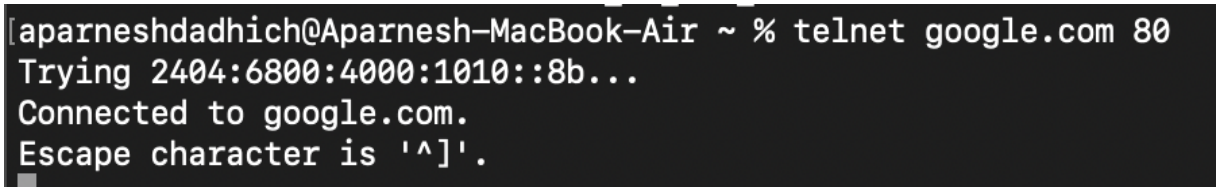
10. TELNET Command

Objective

To test remote connectivity and service availability.

Exercise Questions

1. Connect to a remote host using telnet.



2. Test connectivity to port 80 of a web server.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % telnet google.com 80
Trying 2404:6800:4000:1010::8b...
Connected to google.com.
Escape character is '^]'.
```

3. Explain why telnet is considered insecure.

Telnet is considered insecure because it transmits all data, including usernames and passwords, in plain text. This allows attackers to intercept and read the data using network sniffing, so SSH is preferred for secure remote access.

4. Compare telnet and SSH.

Telnet and SSH are both used for remote access, but SSH is much more secure because it encrypts all data exchanged between the client and server. Telnet sends data in plain text, making it vulnerable to interception, so SSH is the preferred choice for remote administration.

5. Observe server response after connection.



```
aparneshdadhich@Aparnesh-MacBook-Air ~ % telnet google.com 80
Trying 192.178.193.101...
Connected to google.com.
Escape character is '^]'.
GET / HTTP/1.1
Host: google.com

HTTP/1.1 301 Moved Permanently
Location: http://www.google.com/
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-xIqfZkXLjpwZYaMPRbq8Cw' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri http
s://csp.withgoogle.com/csp/gws/other-hp
Date: Tue, 21 Jul 2026 04:01:24 GMT
Expires: Thu, 20 Aug 2026 04:01:24 GMT
Cache-Control: public, max-age=2592000
Server: gws
Content-Length: 219
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN

<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

6. What happens when a port is closed?

If a port is closed, the connection request is rejected and the client receives a Connection refused error. This indicates that no service is listening on that port.

7. Use telnet to manually send HTTP requests.



```
aparneshdadhich@Aparnesh-MacBook-Air ~ % telnet google.com 80
Trying 192.178.193.101...
Connected to google.com.
Escape character is '^]'.
GET / HTTP/1.1
Host: google.com

HTTP/1.1 301 Moved Permanently
Location: http://www.google.com/
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-xIqfZkXLjpwZYaMPRbq8Cw' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri http
s://csp.withgoogle.com/csp/gws/other-hp
Date: Tue, 21 Jul 2026 04:01:24 GMT
Expires: Thu, 20 Aug 2026 04:01:24 GMT
Cache-Control: public, max-age=2592000
Server: gws
Content-Length: 219
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN

<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

8. Explain plaintext transmission in telnet.

Telnet transmits all data, including usernames, passwords, and commands, in plain text without encryption. Anyone intercepting the network traffic can read this information, making Telnet insecure.

9. Identify scenarios where telnet is still useful.

Telnet is still useful for testing network connectivity, checking if a specific port is open, and troubleshooting network services. It is also used in some legacy systems and devices that do not support SSH.

10. Record observations for successful and failed connections.

Successful connection:
The connection is established, and the server displays a welcome message or waits for input, indicating the host and port are reachable.

Failed connection:
The connection fails with an error such as Connection refused or Connection timed out, indicating the port is closed, blocked, or the host is unreachable.

11. SSH Command

Objective

To securely access remote systems.

Exercise Questions

1. Establish SSH connection to a remote host.
2. Explain the role of encryption in SSH.
3. Generate SSH keys and explain their purpose. 4. Compare password authentication and key authentication. 5. Copy files securely using SCP.
6. Explain the role of port 22.
7. What happens when host key verification fails?
8. Compare SSH and Telnet security.
9. Identify common SSH applications in networking.
10. Observe SSH session establishment steps.

12. ARP Command

Objective

To examine Address Resolution Protocol tables.

Exercise Questions

1. Display the ARP cache using arp command.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % arp -a
uniboxadmin.wifi-soft.com (10.21.0.3) at 9c:69:b4:66:44:70 on en0 ifscope [ethernet]
? (10.21.0.18) at 9a:df:cb:4d:38:d4 on en0 ifscope [ethernet]
? (10.21.0.59) at 0:e0:4c:68:5:76 on en0 ifscope [ethernet]
? (10.21.0.75) at 16:6d:9c:a5:6c:50 on en0 ifscope [ethernet]
? (10.21.0.88) at 76:2b:e0:89:52:e1 on en0 ifscope [ethernet]
? (10.21.0.242) at 36:1d:88:5b:ce:86 on en0 ifscope [ethernet]
? (10.21.0.249) at 78:3f:4d:89:dd:7f on en0 ifscope [ethernet]
? (10.21.1.52) at 62:d2:8c:8f:55:2c on en0 ifscope [ethernet]
? (10.21.1.88) at (incomplete) on en0 ifscope [ethernet]
? (10.21.1.124) at 88:1e:5a:7e:dd:16 on en0 ifscope [ethernet]
? (10.21.1.200) at 38:e1:3d:94:ea:20 on en0 ifscope [ethernet]
? (10.21.2.100) at a8:7e:ea:42:59:23 on en0 ifscope [ethernet]
? (10.21.3.86) at ca:64:60:35:50:39 on en0 ifscope [ethernet]
? (10.21.3.87) at a6:3b:b2:c1:23:5d on en0 ifscope [ethernet]
? (10.21.3.113) at 76:b6:4b:17:ae:ef on en0 ifscope [ethernet]
? (10.21.3.122) at 50:bb:b5:48:17:78 on en0 ifscope [ethernet]
? (10.21.3.129) at b2:d6:2b:ba:7e:da on en0 ifscope [ethernet]
? (10.21.3.130) at 86:b7:4d:e3:9:50 on en0 ifscope [ethernet]
? (10.21.3.143) at 52:97:3d:7b:9d:b on en0 ifscope [ethernet]
? (10.21.3.146) at f4:be:ec:ba:96:87 on en0 ifscope [ethernet]
? (10.21.3.187) at 78:fb:d8:f1:6d:50 on en0 ifscope [ethernet]
? (10.21.3.190) at 86:5e:56:f3:30:ef on en0 ifscope [ethernet]
? (10.21.3.206) at 4a:f2:5f:4c:64:28 on en0 ifscope [ethernet]
? (10.21.3.242) at c2:a9:31:bf:5d:99 on en0 ifscope [ethernet]
? (10.21.3.243) at 5a:3a:90:b0:b0:a0 on en0 ifscope [ethernet]
? (10.21.3.250) at 26:28:81:67:2c:11 on en0 ifscope [ethernet]
? (10.21.4.17) at 26:17:7e:26:2d:8 on en0 ifscope [ethernet]
? (10.21.4.32) at 8a:84:93:f7:c1:d0 on en0 ifscope [ethernet]
? (10.21.4.50) at 4a:ac:a3:81:5d:2f on en0 ifscope [ethernet]
? (10.21.4.53) at 78:a7:c7:a3:dc:14 on en0 ifscope [ethernet]
? (10.21.4.69) at 42:a0:18:42:7e:74 on en0 ifscope [ethernet]
? (10.21.4.70) at (incomplete) on en0 ifscope [ethernet]
? (10.21.4.106) at 8a:e6:f2:7b:3a:63 on en0 ifscope [ethernet]
? (10.21.4.121) at a6:45:36:7d:12:2b on en0 ifscope [ethernet]
? (10.21.4.124) at 3c:a6:f6:4:60:cd on en0 ifscope [ethernet]
? (10.21.4.137) at e:a1:5:f4:6c:e8 on en0 ifscope [ethernet]
? (10.21.4.144) at ee:28:6b:bc:dc:be on en0 ifscope [ethernet]
? (10.21.4.207) at f8:7d:76:dd:23:cb on en0 ifscope [ethernet]
? (10.21.5.6) at 26:b8:b0:4:33:47 on en0 ifscope [ethernet]
? (10.21.5.33) at 3e:c6:2b:76:36:15 on en0 ifscope [ethernet]
? (10.21.5.71) at 80:a9:97:48:a3:4d on en0 ifscope [ethernet]
? (10.21.5.73) at 80:a9:97:26:ac:ae on en0 ifscope [ethernet]
? (10.21.5.79) at fe:9:b6:e7:c:d0 on en0 ifscope [ethernet]
? (10.21.5.92) at ce:4c:28:3c:ee:1d on en0 ifscope [ethernet]
? (10.21.5.105) at aa:ef:42:23:8a:71 on en0 ifscope [ethernet]
? (10.21.5.107) at 14:7f:ce:aa:a:79 on en0 ifscope [ethernet]
? (10.21.5.112) at 2c:ca:16:47:e8:b1 on en0 ifscope [ethernet]
? (10.21.5.130) at e6:c4:3e:f7:4d:2b on en0 ifscope [ethernet]
? (10.21.5.155) at 56:b9:a0:f7:62:de on en0 ifscope [ethernet]
```

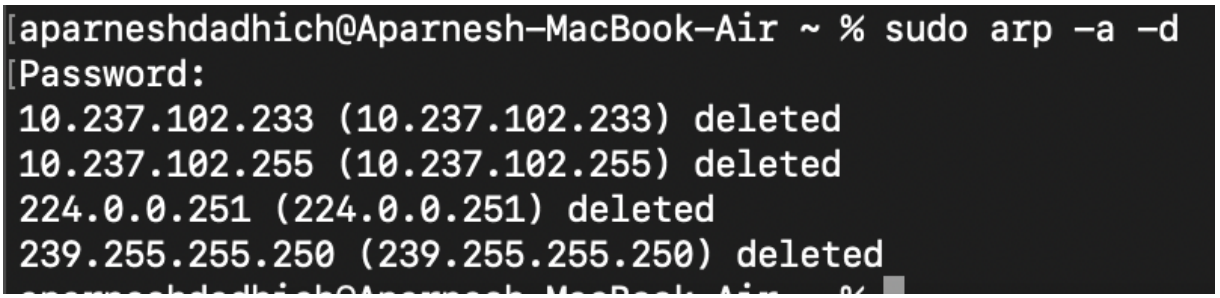
2. Identify IP and MAC address mappings.

   The ARP cache maps each IP address to its corresponding MAC address. Entries marked **(incomplete)** indicate that the MAC address could not be resolved.

3. Explain the role of ARP in LAN communication.

   ARP (Address Resolution Protocol) maps an IP address to its corresponding MAC address on a local network. This allows devices in a LAN to deliver Ethernet frames to the correct destination.

4. Clear the ARP cache and observe changes.



   The ARP cache is cleared, and ARP entries are removed. New entries are automatically created again as the system communicates with devices on the network.

5. Add a static ARP entry.

sudo arp -s <IP_ADDRESS> <MAC_ADDRESS>

6. Differentiate static and dynamic ARP entries.

   Static ARP entries are manually added by the user and remain in the ARP table until removed.

   Dynamic ARP entries are created automatically through ARP requests and are removed after a timeout or when the cache is cleared.

7. Explain ARP request and ARP reply.

   An **ARP request** is a broadcast message sent to all devices on the local network asking for the MAC address of a specific IP address.

   ARP reply is a unicast response from the device with that IP, providing its MAC address.

8. What is Gratuitous ARP?

   Gratuitous ARP is an ARP message sent by a device to announce or update its own IP-to-MAC address mapping without being requested. It is used to detect IP address

conflicts and update ARP caches on other devices.

9. Explain ARP spoofing attacks.

   ARP spoofing is an attack where a malicious device sends fake ARP replies to associate its MAC address with another device's IP address. This allows the attacker to intercept, modify, or block network traffic between devices.

10. Observe changes after pinging a new host.

    After pinging the new host, a new ARP entry for its IP address appears in the ARP cache if it was not already present.

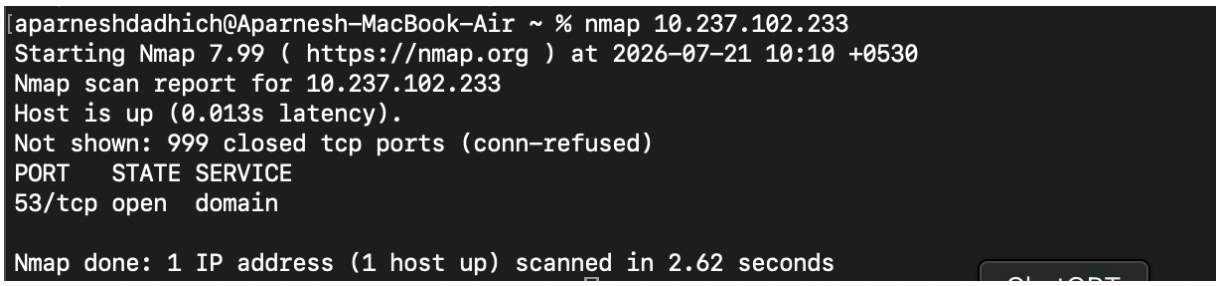
13. NMAP Command

Objective

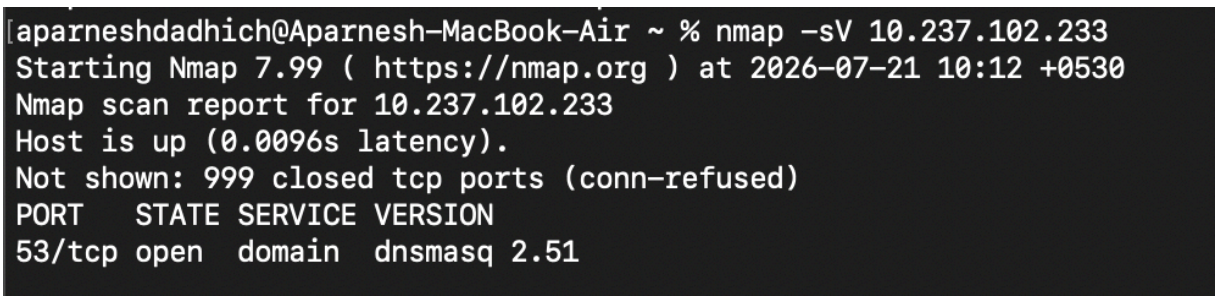
To scan hosts and discover open ports/services.

Exercise Questions

1. Scan a host for open ports.



2. Identify running services using nmap.



3. Perform TCP SYN scan.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % sudo nmap -sS 10.237.102.233
Password:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-21 10:15 +0530
Nmap scan report for 10.237.102.233
Host is up (0.0066s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
53/tcp open  domain
MAC Address: C6:BE:18:80:A0:17 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.99 seconds
```

4. Explain stealth scanning.

Stealth scanning is a technique that gathers information about open ports while minimizing detection. TCP SYN (-sS) scanning is a common stealth scan because it does not complete the full TCP handshake.

5. Compare TCP and UDP scanning.

TCP Scanning	UDP Scanning
Faster and more reliable.	Slower and less reliable.
Uses the TCP protocol.	Uses the UDP protocol.
Receives clear responses from open/closed ports.	Often gets no response from open ports.
Commonly used to identify services.	Used to discover UDP-based services like DNS and DHCP.

6. Detect operating system using nmap.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % sudo nmap -O 10.237.102.233
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-21 10:19 +0530
Nmap scan report for 10.237.102.233
Host is up (0.013s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
53/tcp open  domain
MAC Address: C6:BE:18:80:A0:17 (Unknown)
Device type: phone
Running: Google Android 12.X|10.X|11.X, Linux 5.X|4.X
OS CPE: cpe:/o:google:android:12 cpe:/o:linux:linux_kernel:5.4 cpe:/o:linux:linux_kernel:4 cpe:/o:google:android:10 cpe:/o:google:android:11
OS details: Android 12 (Linux 5.4), Android 10 - 12 (Linux 4.14 - 4.19)
Network Distance: 1 hop
```

7. Scan an entire subnet.

   nmap 10.237.102.0/24

8. Explain the significance of filtered ports.

   Filtered ports are blocked by a firewall or packet filtering device, so Nmap cannot determine whether they are open or closed. They indicate that network security measures are restricting access.

9. Perform version detection scan.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % nmap -sV 10.237.102.233
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-21 10:22 +0530
Nmap scan report for 10.237.102.233
Host is up (0.0070s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT   STATE SERVICE VERSION
53/tcp open  domain  dnsmasq 2.51
```

10. Discuss ethical considerations of port scanning.

Port scanning should only be performed on systems you own or have permission to test. Unauthorized scanning may violate security policies or laws and can be considered malicious activity.

14. GETMAC Command

Objective

To identify MAC addresses of network adapters.

Exercise Questions

1. Display MAC addresses using getmac.

```
C:\Users\premp>getmac

Physical Address    Transport Name
=================== ==========================================================
E8-BF-B8-64-18-BA   \Device\Tcpip_{4E9D8F52-7AA7-49B6-A2E9-8D16081BEFCD}

C:\Users\premp>
```

2. Compare physical and logical addressing.

Physical Address (MAC Address)	Logical Address (IP Address)
Identifies a device on a local network.	Identifies a device across different networks.
Assigned by the manufacturer and usually fixed.	Assigned manually or by DHCP and can change.
Works at the Data Link Layer (Layer 2).	Works at the Network Layer (Layer 3).

3. Identify transport name associated with adapters.

```
C:\Users\premp>getmac /v

Connection Name Network Adapter Physical Address    Transport Name
=============== =============== ================== ============================================================
Wi-Fi           Killer(TM) Wi-F E8-BF-B8-64-18-BA   \Device\Tcpip_{4E9D8F52-7AA7-49B6-A2E9-8D16081BEFCD}

C:\Users\premp>
```

4. Explain uniqueness of MAC addresses.

A MAC address is a unique 48-bit hardware address assigned to a network interface by the manufacturer. It ensures that each device can be uniquely identified on a local network.

5. Compare MAC address and IP address.

MAC Address	IP Address
Permanent hardware address of a network interface.	Logical address assigned to a device on a network.
Works at the Data Link Layer (Layer 2).	Works at the Network Layer (Layer 3).
Used for communication within a local network.	Used for communication across different networks.

6. Explain vendor identification from MAC addresses.

   The first 24 bits of a MAC address, called the OUI (Organizationally Unique Identifier), identify the device manufacturer. The remaining 24 bits are assigned by the manufacturer to uniquely identify each device.

7. Why are MAC addresses important in LANs?

   MAC addresses uniquely identify every network device on a Local Area Network (LAN). They are used by switches to forward data frames to the correct destination, ensuring efficient communication. Since MAC addresses are unique, they help prevent data from being delivered to the wrong device.

8. Compare outputs of getmac and ipconfig.

getmac	**ipconfig**
Displays the MAC (physical) addresses of network adapters.	Displays IP configuration details such as IP address, subnet mask, and default gateway.
Used to identify network hardware.	Used to view and troubleshoot network configuration.

9. Explain MAC spoofing.

   MAC spoofing is the process of changing a device's MAC address to impersonate another device on a network. It is often used for testing, privacy, or bypassing MAC-based access controls, but it can also be misused for unauthorized network access.

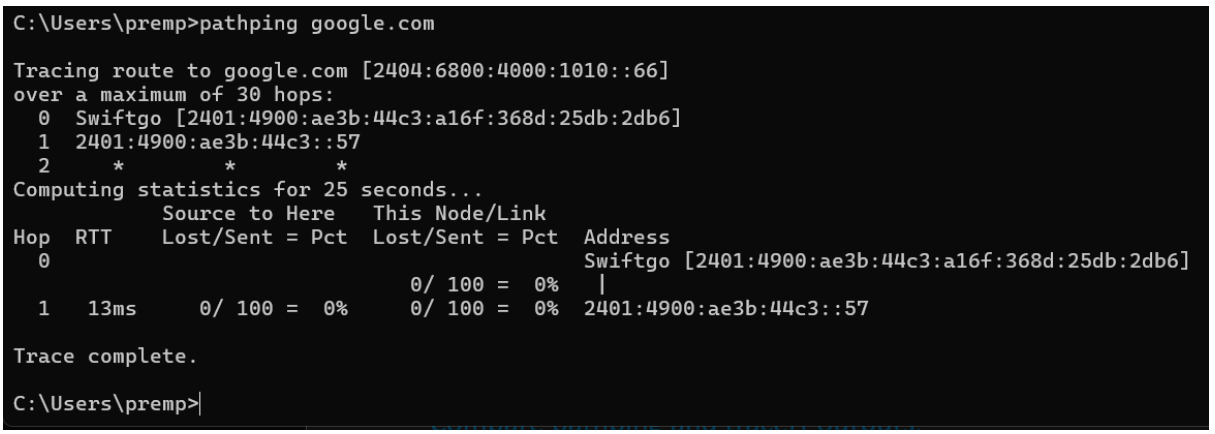
15. PATHPING Command

Objective

To analyze network path and packet loss.

Exercise Questions

1. Execute pathping to a remote server.



```
C:\Users\premp>pathping google.com

Tracing route to google.com [2404:6800:4000:1010::66]
over a maximum of 30 hops:
  0  Swiftgo [2401:4900:ae3b:44c3:a16f:368d:25db:2db6]
  1  2401:4900:ae3b:44c3::57
  2     *        *        *
Computing statistics for 25 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           Swiftgo [2401:4900:ae3b:44c3:a16f:368d:25db:2db6]
                                 0/ 100 =  0%   |
  1   13ms     0/ 100 =  0%     0/ 100 =  0%  2401:4900:ae3b:44c3::57

Trace complete.

C:\Users\premp>
```

2. Compare pathping and tracert outputs.

Pathping	**Tracert**
Shows the route and measures packet loss and latency at each hop.	Shows only the route and response time for each hop.
Takes longer to complete because it collects statistics.	Completes quickly without detailed analysis.
Useful for diagnosing network performance issues.	Useful for identifying the path packets take to the destination.

3. Identify packet loss percentage at each hop.

```
PS C:\Users\jaysh> pathping -4 google.com

Tracing route to google.com [142.251.106.100]
over a maximum of 30 hops:
  0  Tyllenol [10.60.200.126]
  1  10.60.200.73
  2  192.168.17.10
  3  192.168.16.109
  4  192.168.19.21
  5  192.168.19.33
  6  192.168.252.43
  7  125.20.209.146
  8  125.20.209.145
  9  116.119.161.198
 10  72.14.212.48
 11  142.251.225.207
 12  142.250.226.134
 13  192.178.82.111
 14  142.251.254.190
 15     *        *        *
Computing statistics for 350 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           Tyllenol [10.60.200.126]
                                 0/ 100 =  0%   |
  1    9ms     0/ 100 =  0%     0/ 100 =  0%  10.60.200.73
                                 0/ 100 =  0%   |
  2   44ms     0/ 100 =  0%     0/ 100 =  0%  192.168.17.10
                                 0/ 100 =  0%   |
  3   43ms     0/ 100 =  0%     0/ 100 =  0%  192.168.16.109
                                 0/ 100 =  0%   |
  4   60ms     0/ 100 =  0%     0/ 100 =  0%  192.168.19.21
                                 0/ 100 =  0%   |
  5   42ms     0/ 100 =  0%     0/ 100 =  0%  192.168.19.33
                                 0/ 100 =  0%   |
  6   44ms     0/ 100 =  0%     0/ 100 =  0%  192.168.252.43
                                 0/ 100 =  0%   |
  7   49ms     0/ 100 =  0%     0/ 100 =  0%  125.20.209.146
                                 0/ 100 =  0%   |
  8   50ms     0/ 100 =  0%     0/ 100 =  0%  125.20.209.145
                                 0/ 100 =  0%   |
  9   62ms     0/ 100 =  0%     0/ 100 =  0%  116.119.161.198
                                 0/ 100 =  0%   |
 10   63ms     0/ 100 =  0%     0/ 100 =  0%  72.14.212.48
                                 0/ 100 =  0%   |
 11   61ms     0/ 100 =  0%     0/ 100 =  0%  142.251.225.207
                                 0/ 100 =  0%   |
 12   61ms     0/ 100 =  0%     0/ 100 =  0%  142.250.226.134
                                 0/ 100 =  0%   |
 13   64ms     0/ 100 =  0%     0/ 100 =  0%  192.178.82.111
                                 0/ 100 =  0%   |
 14   64ms     0/ 100 =  0%     0/ 100 =  0%  142.251.254.190

Trace complete.
```

4. Explain latency measurements.

   Latency is the time taken for a packet to travel from the source to a destination or network hop, measured in milliseconds (ms). Lower latency indicates a faster and more responsive network connection.

5. Determine the problematic hop in a route.

In my scenario there was no problematic hop.

6. Why does pathping take longer to execute?

   pathping takes longer because it first identifies the route to the destination using a traceroute. It then sends multiple packets to each hop over a period of time to measure packet loss and latency. This extra testing provides detailed network performance statistics, making the command slower than tracert.

7. Compare ping, tracert, and pathping.

Feature	Ping	Tracert	Pathping
Purpose	Tests connectivity to a host.	Shows the route packets take.	Analyzes the route and network performance.
Information Provided	RTT and packet loss.	Network hops and response times.	Hops, latency, and packet loss at each hop.
Execution Time	Fast.	Moderate.	Slow, because it collects detailed statistics.

8. Explain route stability analysis.

Route stability analysis evaluates how consistently data packets travel through the same network path over time. It checks for changes in latency, packet loss, and routing to identify unstable or unreliable network links. Stable routes provide consistent performance with minimal delays and packet loss.

9. Identify congestion using pathping results.

   Network congestion can be identified in pathping results by high latency and increased packet loss at one or more hops. If these values suddenly increase compared to previous hops, it indicates congestion or overload at that part of the network.

10. Record observations for a long-distance destination.

    For a long-distance destination, the route usually contains more network hops and higher latency than a nearby destination. Packet loss is typically low on a stable network, but latency gradually increases as the distance to the destination grows.

16. SS Command

Objective

To investigate socket statistics in Linux.

Exercise Questions

1. Display active TCP connections using ss.

```
mithrix@Tyllenol:~$ ss -ta
State       Recv-Q  Send-Q     Local Address:Port       Peer Address:Port  Process
LISTEN      0       4096       127.0.0.53%lo:domain          0.0.0.0:*
LISTEN      0       4096          127.0.0.54:domain          0.0.0.0:*
LISTEN      0       1000      10.255.255.254:domain          0.0.0.0:*
TIME-WAIT   0       0         172.19.118.126:37454      23.61.247.19:http
```

2. Compare ss and netstat commands.

ss	netstat
Faster and more efficient.	Slower, especially on systems with many connections.
Uses kernel socket statistics directly.	Reads information from /proc and other sources.
Modern replacement for netstat on Linux.	Older command; deprecated on many Linux distributions.

3. Display listening ports using ss.

```
mithrix@Tyllenol:~$ ss -lt
State    Recv-Q   Send-Q      Local Address:Port       Peer Address:Port  Process
LISTEN   0        4096        127.0.0.53%lo:domain          0.0.0.0:*
LISTEN   0        4096           127.0.0.54:domain          0.0.0.0:*
LISTEN   0        1000       10.255.255.254:domain          0.0.0.0:*
```

4. Filter connections by protocol.

```
mithrix@Tyllenol:~$ ss -t
State    Recv-Q    Send-Q      Local Address:Port      Peer Address:Port    Process
mithrix@Tyllenol:~$ ss -u
Recv-Q     Send-Q         Local Address:Port         Peer Address:Port     Process
```

5. Identify process associated with a socket.

```
mithrix@Tyllenol:~$ sudo ss -tulpn
Netid    State     Recv-Q    Send-Q       Local Address:Port      Peer Address:Port    Process
udp      UNCONN    0         0               127.0.0.54:53            0.0.0.0:*        users:(("systemd-resolve",pid=131,fd=16))
udp      UNCONN    0         0            127.0.0.53%lo:53            0.0.0.0:*        users:(("systemd-resolve",pid=131,fd=14))
udp      UNCONN    0         0           10.255.255.254:53            0.0.0.0:*
udp      UNCONN    0         0                127.0.0.1:323           0.0.0.0:*
udp      UNCONN    0         0                    [::1]:323              [::]:*
tcp      LISTEN    0         4096         127.0.0.53%lo:53            0.0.0.0:*        users:(("systemd-resolve",pid=131,fd=15))
tcp      LISTEN    0         4096            127.0.0.54:53            0.0.0.0:*        users:(("systemd-resolve",pid=131,fd=17))
tcp      LISTEN    0         1000        10.255.255.254:53            0.0.0.0:*
```

6. Explain socket states observed.

LISTEN – The socket is waiting for incoming connection requests on a specific port. It is commonly used by servers before a client connects.

ESTABLISHED – The connection between the client and server is active, allowing data to be sent and received.

TIME_WAIT – The connection has been closed, but the system waits for a short period to ensure delayed packets do not affect future connections.

CLOSE_WAIT – The remote host has closed the connection, but the local application has not yet closed its side of the socket.

7. Display summary statistics using ss.

```
mithrix@Tyllenol:~$ ss -s
Total: 182
TCP:   3 (estab 0, closed 0, orphaned 0, timewait 0)

Transport Total     IP        IPv6
RAW       0         0         0
UDP       5         4         1
TCP       3         3         0
INET      8         7         1
FRAG      0         0         0
```

8. Explain advantages of ss over netstat.

The ss command is faster and more efficient than netstat because it retrieves socket information directly from the Linux kernel. It provides more detailed information about active connections, including process details and socket statistics. Since it uses fewer system resources and offers better performance, ss has become the preferred networking utility on modern Linux systems.

9. Observe IPv4 and IPv6 socket entries.

```
mithrix@Tyllenol:~$ ss -tulpn
Netid  State   Recv-Q  Send-Q    Local Address:Port   Peer Address:Port Process
udp    UNCONN  0       0           127.0.0.54:53          0.0.0.0:*
udp    UNCONN  0       0         127.0.0.53%lo:53         0.0.0.0:*
udp    UNCONN  0       0        10.255.255.254:53         0.0.0.0:*
udp    UNCONN  0       0             127.0.0.1:323        0.0.0.0:*
udp    UNCONN  0       0                 [::1]:323           [::]:*
tcp    LISTEN  0       4096      127.0.0.53%lo:53         0.0.0.0:*
tcp    LISTEN  0       4096         127.0.0.54:53         0.0.0.0:*
tcp    LISTEN  0       1000     10.255.255.254:53         0.0.0.0:*
```

10. Identify high network usage processes.

The top 2 TCP processes above are using the most network.

17. ROUTE ADD Command

Objective
To add static routes manually.

Exercise Questions

1. Add a static route to a network.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % sudo route -n add -net 192.168.2.0/24 192.168.1.1
Password:
add net 192.168.2.0: gateway 192.168.1.1
```

2. Verify the route addition using routing table.

```
192.168.2          192.168.1.1        UGSc                   en0
```

3. Delete the added route.

```
aparneshdadhich@Aparnesh-MacBook-Air ~ % sudo route -n delete -net 192.168.2.0/24
delete net 192.168.2.0
```

4. Explain gateway selection in route add.

The gateway specified in the route add command acts as the next-hop router that forwards packets toward the destination network. It must be reachable from the local system and belong to the same network as the source host. Choosing the correct gateway ensures that data is routed efficiently to its destination.

5. Compare persistent and temporary routes.

Persistent Route	Temporary Route
Remains after the system restarts.	Removed automatically after a system restart.
Used for permanent network configurations.	Used for temporary or testing purposes.

In Windows, created using the `-p` option with `route add`.	Created using `route add` without the `-p` option.

6. Add a route with metric value.

```
PS C:\Users\jaysh> route add 192.168.50.0 mask 255.255.255.0 192.168.1.1 metric 5
 OK!
```

7. Observe packet forwarding after route addition.

After adding the route, packets are forwarded through the specified gateway instead of the default route. The routing table is checked to select the matching route for the destination network. This ensures that traffic follows the newly configured path.

8. What errors occur with invalid gateways?

If an invalid gateway is specified, the route may fail to be added or packets cannot be forwarded to the destination. The system may display errors. As a result, communication with the destination network fails.

9. Explain practical applications of static routes.

Static routes are used in small networks where the routing path rarely changes. They provide predictable routing, improve security by controlling traffic paths, and reduce routing overhead. They are also commonly used for connecting branch offices or reaching specific networks.

10. Differentiate route add and dynamic routing protocols.

route add (Static Routing)	**Dynamic Routing Protocols**
Routes are added manually by the administrator.	Routes are learned and updated automatically.
Best for small, stable networks.	Best for large and changing networks.
Does not adapt to network changes automatically.	Automatically adjusts routes when the network changes.